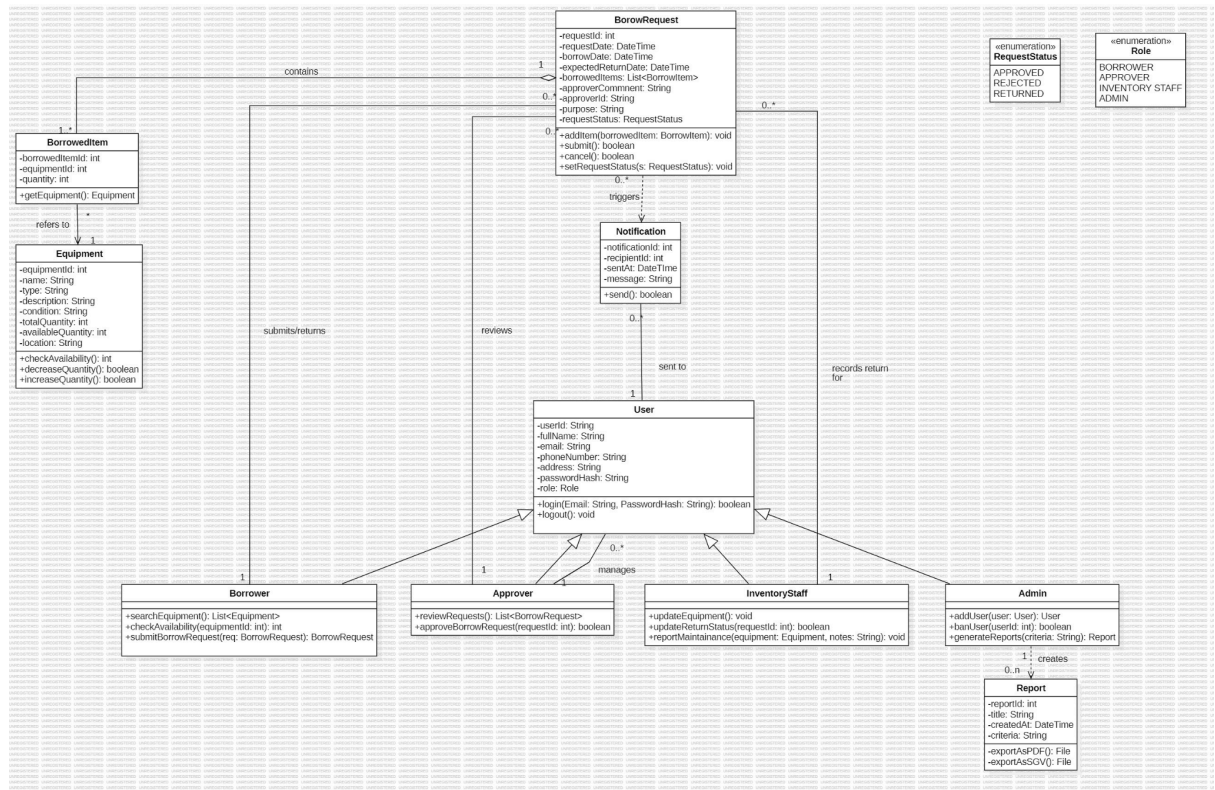


LAB 4

1. UML Class Diagram



If the background is bothersome, please click the following drive link to see pdf (clearer background):

<https://drive.google.com/file/d/11XALo4gwPWCF8bgMbLzj4pA7NNqsQQr-/view?usp=sharing>

2. Table listing all classes, attributes, and methods.

Class Name	Attributes	Methods
BorrowRequest	- requestId: String - requestDate: Date - status: String - purpose: String - borrower: Borrower - approver: Approver - inventoryStaff: InventoryStaff - borrowedItems: List<BorrowedItem>	- approveRequest() - rejectRequest() - returnItems() - updateStatus() - getRequestDetails()

Notification	• notificationId: String • message: String • date: Date • user: User	• sendNotification() • getNotificationDetails()
User	• userId: String • name: String • email: String • password: String • role: String	• login() • logout() • updateProfile() • getUserDetails()
Borrower	<i>(inherits User)</i>	• createBorrowRequest() • viewBorrowRequest() • returnItems()
Approver	<i>(inherits User)</i>	• approveRequest() • rejectRequest() • viewRequests()
InventoryStaff	<i>(inherits User)</i>	• manageInventory() • updateItemStatus() • viewBorrowedItems()
Admin	<i>(inherits User)</i>	• manageUsers() • generateReports() • viewSystemLogs()
Equipment	• equipmentId: String • name: String • description: String • quantity: int • status: String	• addEquipment() • updateEquipment() • deleteEquipment() • getEquipmentDetails()
BorrowedItem	• itemId: String • equipment: Equipment • quantity: int • borrowRequest: BorrowRequest	• updateItemStatus() • getItemDetails()
Report	• reportId: String • title: String • content: String • date: Date • generatedBy: Admin	• generateReport() • viewReport() • deleteReport()

3. Notes on inheritance and associations.

a. Inheritance Notes

Subclass	Inherits From	Purpose
Borrower	User	Represents users who request to borrow equipment.
Approver	User	Represents users who approve or reject borrow requests.
InventoryStaff	User	Manages inventory and tracks borrowed items.
Admin	User	Oversees system operations, user management, and report generation.

b. Association Notes

Class A	Class B	Type of Association	Description
BorrowRequest	Borrower	Aggregation	Each request is created by a borrower.
BorrowRequest	Approver	Aggregation	Each request is reviewed by an approver.
BorrowRequest	InventoryStaff	Aggregation	Inventory staff handle the logistics of the request.
BorrowRequest	BorrowedItem	Composition	A request contains one or more borrowed items.
BorrowedItem	Equipment	Association	Each borrowed item references a specific piece of equipment.
BorrowedItem	BorrowRequest	Association	Links the item back to the request it belongs to.
Notification	User	Association	Notifications are sent to users.
Report	Admin	Association	Reports are generated by admins.